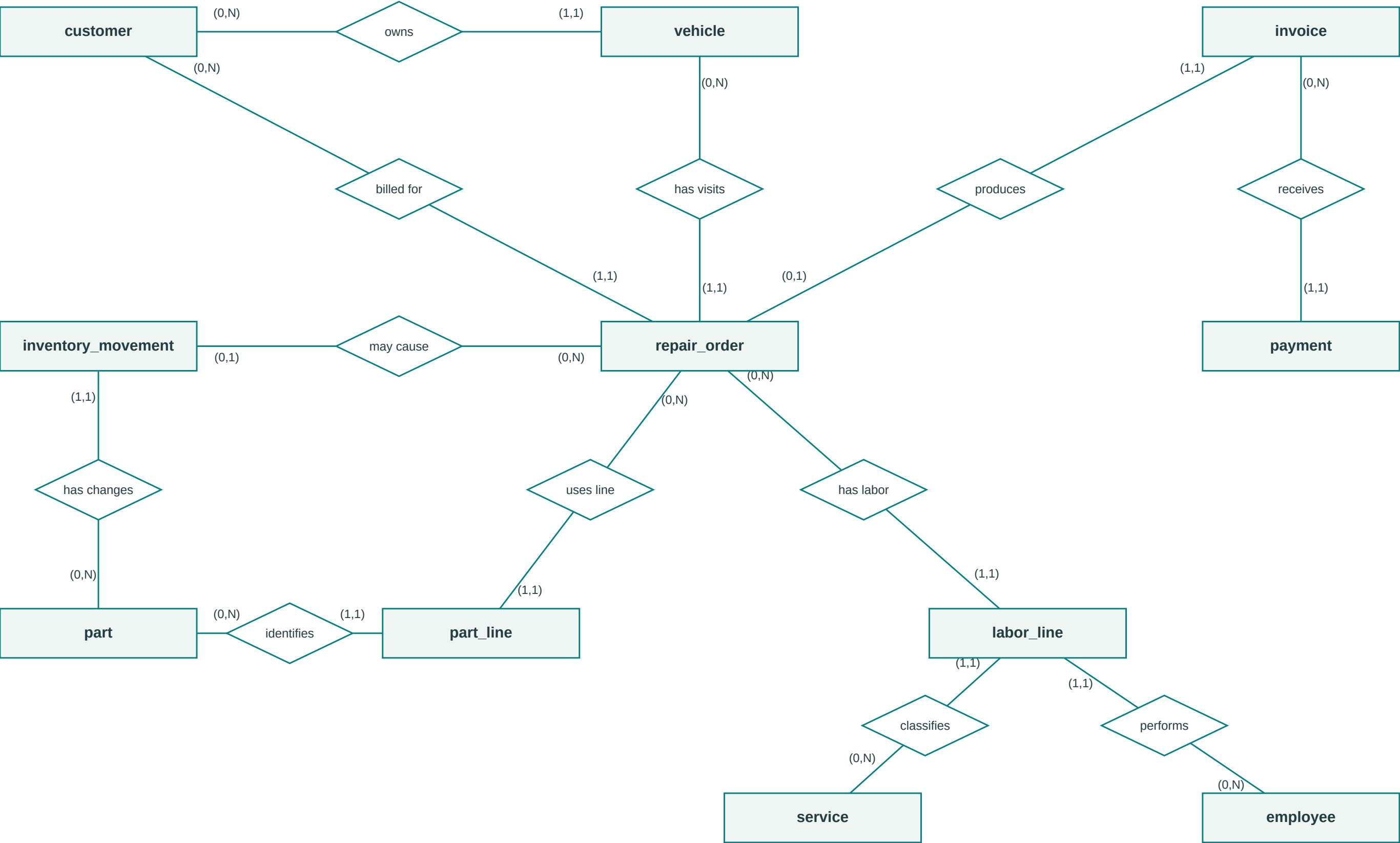


Leo's Auto Chen ERD

Sheet 1: all 11 entities and all 12 relationships. Attribute ovals are on sheet 2; repeated names mean the same entity.

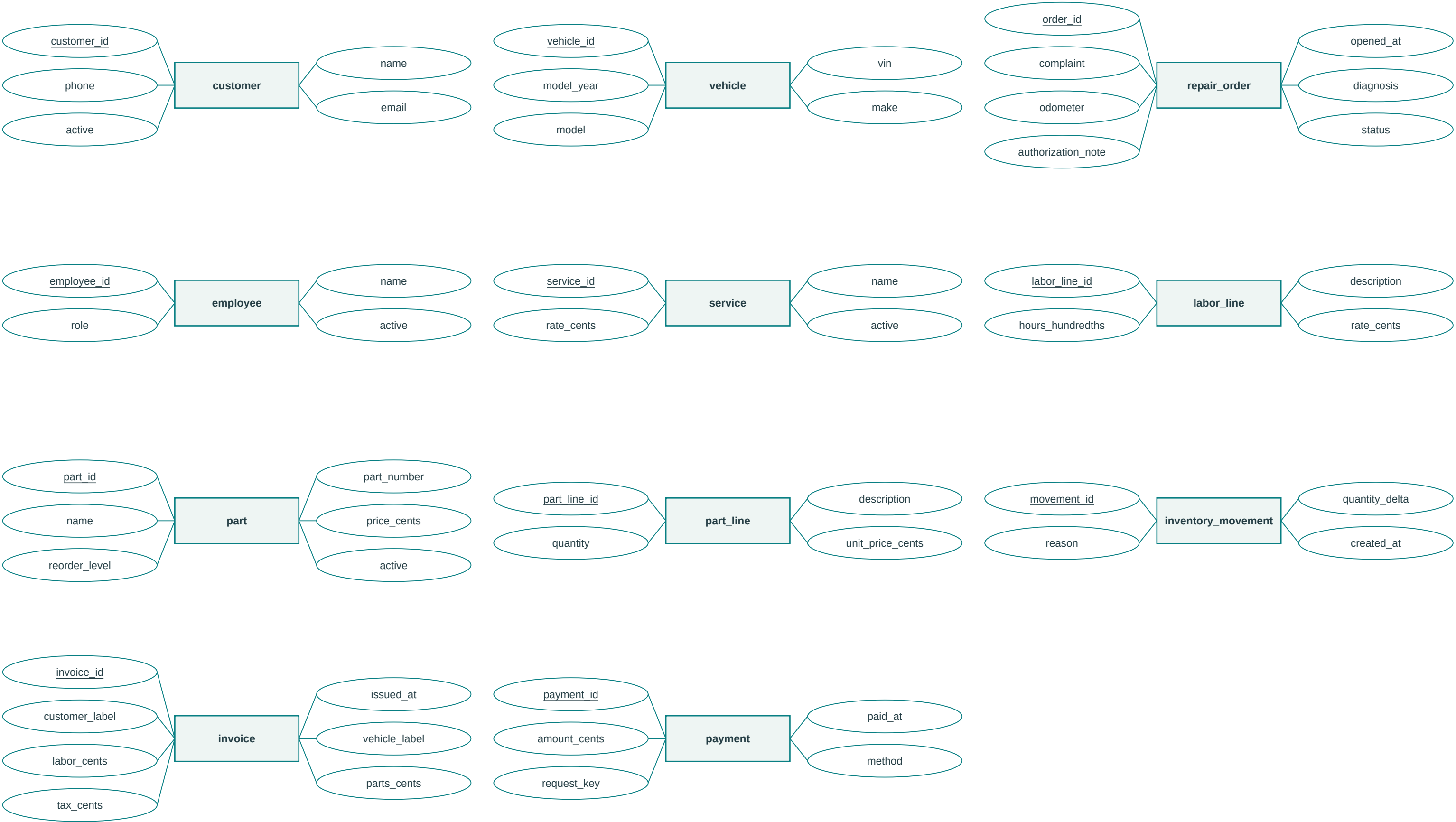
Rectangles = things stored. Diamonds = relationships. (0,N) = zero or many; (1,1) = exactly one; (0,1) = optional one.

Read the label beside an entity as its participation: one vehicle has (1,1) owner; one customer can own (0,N) vehicles.



Chen attributes

Sheet 2: all non-FK fields in the sample schema. Underlined IDs are primary keys. Sheet 1 relationships become the foreign keys.



Stock and outstanding balance are calculated views, not extra entities. Fields on this sheet map one-for-one to the schema on pages 3-4.

Matching schema

PK = unique row ID. FK = link to another table. Every table has one integer PK. The ERD relationships supply the FKs below.

customer - Contact details for a customer.

PK customer_id, name, phone, email, active

vehicle - A vehicle and its current customer.

PK vehicle_id, FK customer_id, vin, model_year, make, model

customer_id links to customer.customer_id

Also unique: vin.

repair_order - One visit; bill-to customer is recorded separately from current ownership.

PK order_id, FK vehicle_id, FK bill_to_customer_id, opened_at, complaint, diagnosis, odometer, status, authorization_note

bill_to_customer_id links to customer.customer_id; vehicle_id links to vehicle.vehicle_id

employee - Staff; only an active mechanic can be selected for a labor line.

PK employee_id, name, role, active

service - Catalog of labor types and current rates.

PK service_id, name, rate_cents, active

labor_line - A service performed on a job, by one mechanic, at the rate charged.

PK labor_line_id, FK order_id, FK service_id, FK mechanic_id, description, hours_hundredths, rate_cents

mechanic_id links to employee.employee_id; service_id links to service.service_id; order_id links to repair_order.order_id

Converting Chen to tables

An entity becomes a table; its ovals become columns. For one-to-many links, put the FK on the many side. For the optional one-to-one order/invoice link, invoice.order_id is a unique FK. LaborLine and PartLine are separate line entities so each use can store its own quantity, hours and price.

Schema and simple rules

part - Part catalog; stock is calculated from inventory movements.

PK part_id, part_number, name, price_cents, reorder_level, active

Also unique: part_number.

part_line - Parts used on a job, at the price charged.

PK part_line_id, FK order_id, FK part_id, description, quantity, unit_price_cents

part_id links to part.part_id; order_id links to repair_order.order_id

inventory_movement - Each stock increase or decrease; a repair link is optional.

PK movement_id, FK part_id, FK order_id, quantity_delta, reason, created_at

order_id links to repair_order.order_id (optional); part_id links to part.part_id

invoice - One final bill per order, with labels and amounts preserved.

PK invoice_id, FK order_id, issued_at, customer_label, vehicle_label, labor_cents, parts_cents, tax_cents

order_id links to repair_order.order_id

Also unique: order_id.

payment - One payment toward an invoice; several partial payments are allowed.

PK payment_id, FK invoice_id, paid_at, amount_cents, method, request_key

invoice_id links to invoice.invoice_id

Also unique: request_key.

Rules already in the sample

Money is stored as integer cents; hours as hundredths. Stock cannot go below zero. Payments must be positive and cannot exceed the balance. VINs cannot repeat. Only authorized, in-progress orders accept work lines. Final orders and issued bills are locked.

Confirm before keeping these choices

One current customer per vehicle; one mechanic per labor line; at most one final invoice per visit. No deposits before invoicing, refunds, invoice corrections or shared vehicle ownership yet. Tax is a manually entered demo amount. Change the model if the interview requires it.

Why separate tables?

Customer details, service rates and part prices each have one home. Job lines preserve the rate actually charged; invoice labels preserve the billed details. These are historical facts, not copies to keep synchronized. Stock = sum of movements; balance = invoice total minus payments.